**Mini Project report on**

# SENSUS CRUOR

**Submitted by**


**ALEENA PAUL**

**FIT22EC018**


**Focus on Excellence**


**Department of Electronics & Communication Engineering**

**FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT)®**
**Angamaly-683577, Ernakulam**

Affiliated to

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**FEDERAL INSTITUTE OF SCIENCE AND TECHNOLOGY (FISAT) ®**

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**Focus on Excellence**

# CERTIFICATE

This is to certify that the Mini project report titled **SENSUS CRUOR** submitted by **ALEENA PAUL**, towards partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in Electronics and Communication Engineering is a record of bonafide work carried out by her during the academic year 2024-2025.

Project Guide

Head of the Department

Internal Examiner

External Examiner

Place: Mookkannoor

Date:

## ACKNOWLEDGMENT

I have taken efforts in this project. However, it would not have been possible without the kind support and help of many individuals and organizations. I would like to extend my sincere thanks to all of them.

I am highly indebted to **Asst. Prof Anoop E G** for his guidance and **Asst. Prof Nimmy M Philip** for her co guidance and their constant supervision as well as for providing necessary information regarding the project.

I would like to express my gratitude towards my parents & batch-mates for their kind cooperation and encouragement which helped me in the completion of this project. I would like to express my special gratitude and thanks to my mentors for giving me such attention and time and helping me throughout the duration of the project.

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My thanks and appreciation also goes to my colleagues in developing the project and to people who have willingly helped me out with their abilities.

# ABSTRACT

*Sensus Cruor is a smart menstruation detection system designed to support differently-abled individuals, particularly those with ASD, who may struggle with recognizing and responding to menstruation. This system integrates an wet sensor and an IoT-based alert mechanism to detect menstrual blood and send real-time notifications to caregivers via Telegram, ensuring timely assistance while preserving the individual's privacy and dignity. Unlike traditional methods that rely on manual checks, Sensus Cruor provides an autonomous and reliable solution, enhancing menstrual hygiene management and reducing caregiver dependency. Utilizing ESP8266, sensor technology, and real-time alerts, this innovation offers a practical, hygienic, and supportive approach to improving the quality of life for individuals with special needs.*

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# ABBREVIATIONS

**ASD: AUTISM SPECTRUM DISORDER**

**API: APPLICATION PROGRAMMING INTERFACE**

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# Chapter 1

# INTRODUCTION

Sensus Cruor is a Latin phrase meaning "Sense of Blood" representing an advanced menstruation detection system designed to assist differently-abled individuals, particularly those with ASD. Many individuals with ASD may struggle with recognizing and managing menstruation, leading to hygiene concerns, discomfort, and dependency on caregivers for assistance. Delayed recognition of menstruation can cause distress and disrupt daily activities, highlighting the need for an automated, efficient solution.

To address these challenges, Sensus Cruor integrates a wet sensor with an IoT-based alert mechanism to detect menstrual blood and send real-time notifications to caregivers via Telegram. The wet sensor, designed to identify the presence of fluid based on resistance changes, enables accurate detection of menstrual blood without requiring intrusive monitoring. Once detected, the system instantly transmits alerts, ensuring caregivers can provide timely support while maintaining the individual's privacy and dignity. Unlike traditional methods that rely on manual checks, this system minimizes discomfort, reduces dependency, and enhances personal hygiene management.

The system is powered by the ESP8266 microcontroller, enabling seamless wireless communication and real-time data transmission. By leveraging sensor technology, IoT connectivity, and automated notifications, Sensus Cruor offers a proactive and reliable approach to menstrual hygiene management. This innovation enhances caregiver efficiency, fosters self-reliance among differently-abled individuals, and promotes a more inclusive and dignified approach to menstrual health.

# Chapter 2
# LITERATURE REVIEW

## R. Steward, L. Crane, E. M. Roy, A. Remington, and E. Pellicano, "“Life is much more difficult to manage during periods”: Autistic experiences of menstruation," *Autism*, vol. 22, no. 7, pp. 839-846, 2018.

### Challenges of Menstruation for Autistic Individuals

*Sensory and Emotional Difficulties*

- Increased *pain sensitivity*, sensory overload, and discomfort with products.
- Heightened sensitivity to *smells, noise, and touch*, leading to meltdowns.
- *Emotional dysregulation* and increased *anxiety* or shutdowns

*Practical and Social Challenges*

- Difficulty *managing hygiene* and tracking cycles due to executive dysfunction.
- Trouble *communicating needs* (e.g., asking for breaks, discussing pain).
- Many feel *unprepared* for their first period due to lack of accessible information.

### Recommendations for Better Support

*Education and Awareness*

- Provide *clear guides* on menstrual care.
- Educate *parents, teachers, and caregivers* about menstruation challenges.
- Increase *research* and *awareness* in healthcare.

*Practical Support*

- Offer *sensory-friendly products*.
- Teach *cycle tracking* and symptom recognition.
- Consider *medical interventions* (e.g., hormonal treatments) for severe cases.

# V. Gaubert, H. Gidik, and V. Koncar, "Boxer underwear incorporating textile moisture sensor to prevent nocturnal enuresis," *Sensors*, vol. 20, no. 12, p. 3487, 2020.

**Introduction**

- Nocturnal Enuresis (NE) affects 15% of children over five.
- Current bedwetting alarms are bulky and uncomfortable.
- A seamless textile moisture sensor is integrated into boxer underwear for improved comfort and better detection

**Key Features**

- Textile sensor made from conductive yarns (silver/stainless steel).
- Seamless integration using circular knitting.
- Miniaturized electronics with alarm to wake the child upon detection.

**Sensor Design & Materials**

- Cotton/elastane substrate for comfort.

Various sensor patterns tested for optimal detection

**Experimental Evaluation**

- Leakage detection speed with artificial urine.
- Durability tests: machine-wash (20 cycles), corrosion resistance, performance under bending/stretching.

**Results & Findings**

- Silver-plated yarn showed fastest, most reliable detection.
- Stainless steel required more liquid for detection.
- Sensor design impacted speed; tight comb design performed best.
- Washing/corrosion reduced conductivity but silver

- remained effective.
- Child-friendly design with snap-on module for easy removal.

**Conclusion & Impact**

- Comfortable, effective solution for enuretic children.
- Seamless integration improves acceptance.
- Potential for mass production and commercial use.
- Can be adapted for other incontinence applications.

# H. Fuketa, K. Yoshioka, T. Yokota, W. Yukita, M. Koizumi, M. Sekino, T. Sekitani, M. Takamiya, T. Someya, and T. Sakurai, "Organic-transistor-based 2kV ESD-tolerant flexible wet sensor sheet for biomedical applications with wireless power and data transmission using 13.56MHz magnetic resonance," *J. Micromech. Microeng.*, 2018

## Introduction

- Flexible wet sensor sheet (FWSS) for biomedical use.
- Detects urination in diapers for elderly care.
- Uses wireless power and data transmission via 13.56MHz magnetic resonance.

## Key Features

- Thin, flexible, and disposable for hygiene and comfort.
- Wireless operation powered by a nearby reader.
- Organic transistors on polyimide film convert resistance changes into frequency signals.

## Results & Findings

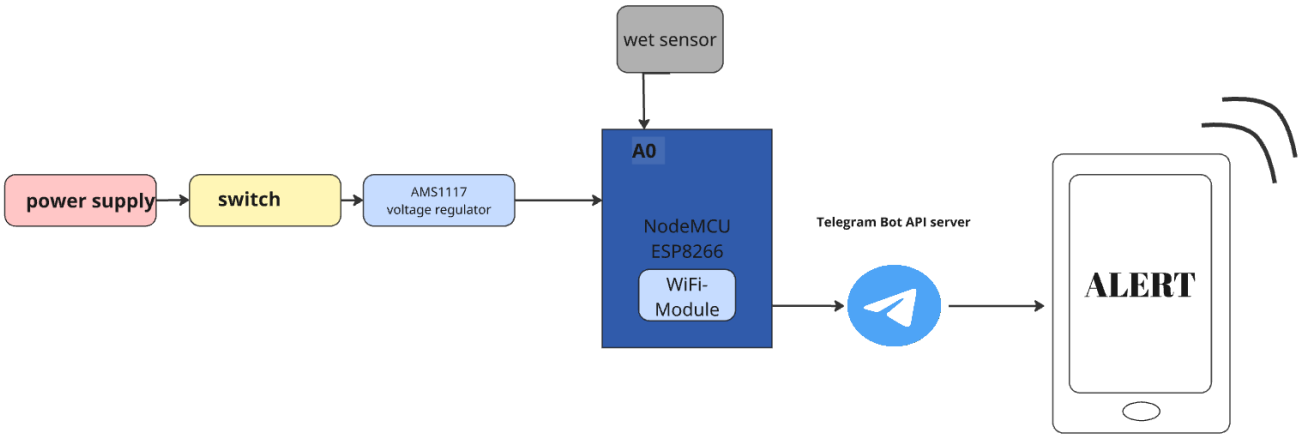
- Successful wireless power transfer with flexible coil.
- Efficient power usage with AAC.
- Durable with ESD protection.
- Urination detection using frequency shifts in RC oscillators.

## Conclusion & Impact

- Innovative use of organic transistors in biomedical sensing.
- Enables real-time health monitoring.
- Potential for wearable and flexible electronics.

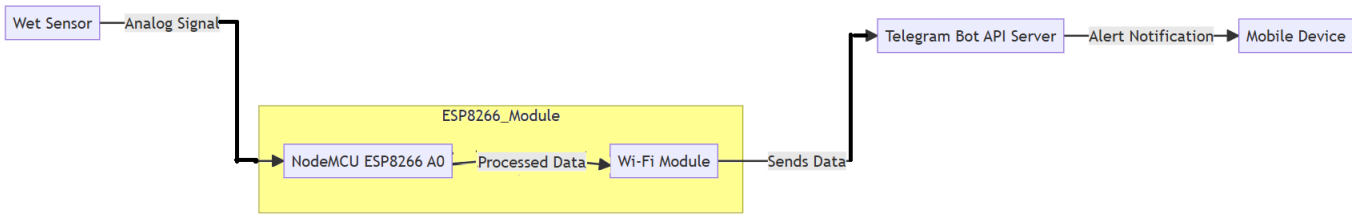
# Chapter 3

# BLOCK DIAGRAM AND WORK FLOW




## 3.1 Block Diagram

The system operates by detecting the presence of liquid using a wet sensor, which measures resistance changes when it comes into contact with blood or urine. This sensor is connected to the A0 pin of the NodeMCU ESP8266, where the microcontroller processes the signal and determines if the measured resistance falls below a predefined threshold, confirming liquid detection. The power supply, controlled by a switch, delivers energy to the AMS1117 voltage regulator, ensuring a stable 3.3V supply to the ESP8266. Once the microcontroller detects liquid, the integrated Wi-Fi module connects to the internet and transmits an alert via the Telegram Bot API. The API server then forwards the notification to a designated Telegram chat or group, instantly informing caregivers or guardians through their mobile devices. This real-time alert system ensures prompt assistance and is particularly beneficial for individuals with autism or those requiring health monitoring, offering a reliable and automated menstrual or hygiene management solution.




## 3.2 flow chart

This flowchart represents the working of a Wet Sensor-Based Menstruation Blood Detection System using ESP8266 and Telegram Bot API for alert notifications. Here's how it works:

1. Wet Sensor:
   - Detects the presence of blood by sensing changes in resistance.
   - Outputs an analog signal.
2. NodeMCU ESP8266 (A0 Pin):
   - Reads the analog signal from the wet sensor.
   - Processes the data to determine if blood is detected.
3. Wi-Fi Module (Part of ESP8266):
   - Sends processed data over the internet.
   - Communicates with a Telegram Bot API Server.
4. Telegram Bot API Server:
   - Receives data from the ESP8266.
   - If blood is detected, it triggers an alert notification.
5. Mobile Device:
   - Receives an alert notification via Telegram.
   - Notifies the caretaker or guardian about the detected event.

Summary:

When the wet sensor detects blood, the ESP8266 processes the data and sends an alert notification through Telegram to a mobile device. This ensures timely alerts for caretakers, helping in managing

menstruation for autistic individuals.

# Chapter 4

# COMPONENTS

The components required to implement the hardware of sensus cruor are described below.

| Sl No | COMPONENTS | QUANTITY |
|---|---|---|
| 4.1 | ESP8266 (NODEMCU) | 1 |
| 4.2 | WET SENSOR | 1 |
| 4.3 | LI-ION BATTERY(3.7V)<br>TP4056 MODULE | 1<br>1 |
| 4.4 | AMS1117 (3.3V) | 1 |
| 4.5 | SLIDE SWITCH | 1 |

TABLE 1: The components required to implement the hardware of Sensus cruor.

## 4.1 ESP8266(NODEMCU)

ESP8266 is a popular and widely used open-source Wi-Fi microcontroller module that has transformed IoT applications and smart device development. Developed by Espressif Systems, it is based on a low-power 32-bit Tensilica LX106 processor. The ESP8266 is designed for seamless wireless communication, making it ideal for IoT projects, automation, and embedded systems.

The key features of ESP8266 are built-in Wi-Fi connectivity, low power consumption, GPIO support, cost-effectiveness, and compatibility with various development platforms like Arduino IDE, MicroPython, and NodeMCU firmware.

## 4.2 WET SENSOR

A wet sensor is a type of moisture detection sensor designed to identify the presence of liquid, including water and other fluids, based on changes in electrical resistance or capacitance. It is

commonly used in applications such as leakage detection, environmental monitoring, and automation systems. In healthcare and hygiene-related projects, wet sensors can detect bodily fluids, making them useful for smart hygiene monitoring solutions. Wet sensors are compact, cost-effective, and easy to integrate with microcontrollers for real-time alerts and automated responses.

## 4.3 LI-ION BATTERY AND TP4056 MODULE

Li-Ion (Lithium-Ion) batteries are rechargeable energy storage devices widely used in portable electronics and IoT applications due to their high energy density, lightweight design, and long cycle life. They provide stable voltage output and efficient power delivery, making them suitable for embedded systems and battery-operated projects.

The TP4056 module is a compact and efficient lithium-ion battery charging and protection circuit. It features built-in overcharge, over-discharge, and short-circuit protection, ensuring safe and reliable charging of Li-Ion batteries. The module uses a micro-USB or Type-C input for charging and provides a regulated output, making it a convenient solution for managing battery-powered applications.

## 4.4 AMS1117(3.3V)

AMS1117-3.3V is a linear voltage regulator designed to provide a stable 3.3V output from a higher voltage source, typically 5V or more. It is widely used in embedded systems, IoT devices, and microcontroller-based circuits to power components that require a consistent 3.3V supply. The regulator offers up to 800mA of output current and features built-in thermal shutdown and current-limiting protection, ensuring safe and efficient operation. Due to its compact size and ease of integration, the AMS1117-3.3V is a popular choice for low-power electronics.

## 4.5 SLIDE SWITCH

A slide switch is a type of mechanical switch that allows users to open or close an electrical circuit by sliding a lever back and forth. In this project, the switch is placed between the ams1117 voltage regulator and the battery module to control the power supply. It enables users to manually turn the voltage regulator on or off, preventing unnecessary battery drain and ensuring efficient power management. When the switch is in the on position, it connects the battery module to the ams1117, supplying a regulated 3.3v output to the esp8266 and other components. When turned off, it disconnects the power, effectively shutting down the regulated output. Slide switches come in various configurations, such as spst (single pole single throw) for simple on/off control, spdt (single pole double throw) for toggling between two outputs, and dpdt (double pole double throw) for controlling two circuits simultaneously. These switches are compact, reliable, and easy to integrate into various electronic projects, including embedded systems, power regulation, and automation applications.

# Chapter 5
# SOFTWARE

The software used for implementing the sensus cruor are described below

## 5.1 ARDUINO IDE

Arduino IDE is a free and open-source development environment used for writing, compiling, and uploading code to Arduino boards and compatible microcontrollers like esp8266. It provides a simple and user-friendly interface, making it accessible for beginners and professionals alike. The ide supports the c and c++ programming languages and includes built-in libraries for various sensors and modules. It features a serial monitor for debugging, code compilation, and an extensive community-driven support system. The key features of Arduino ide are open source and cross-platform compatibility, built-in libraries, a serial monitor, board and port management, syntax highlighting, offline programming, and integration with additional plugins and extensions.

## 5.2 TELEGRAM BOT

Using a Telegram bot in this project involves creating a bot through the Telegram Bot API, integrating it with the esp8266, and setting up a communication link to send real-time notifications. The Telegram bot serves as an automated messaging system that alerts caregivers when the wet sensor detects menstrual blood. This ensures immediate assistance while maintaining privacy and dignity for individuals with autism spectrum disorder. The bot can receive and process commands, send updates, and provide a secure and reliable communication channel. Overall, the Telegram bot simplifies remote monitoring by offering instant alerts, easy integration, and accessibility across multiple devices, making it an efficient tool for real-time notifications in IoT-based healthcare applications.

# Chapter 6

# COST OF PROJECT

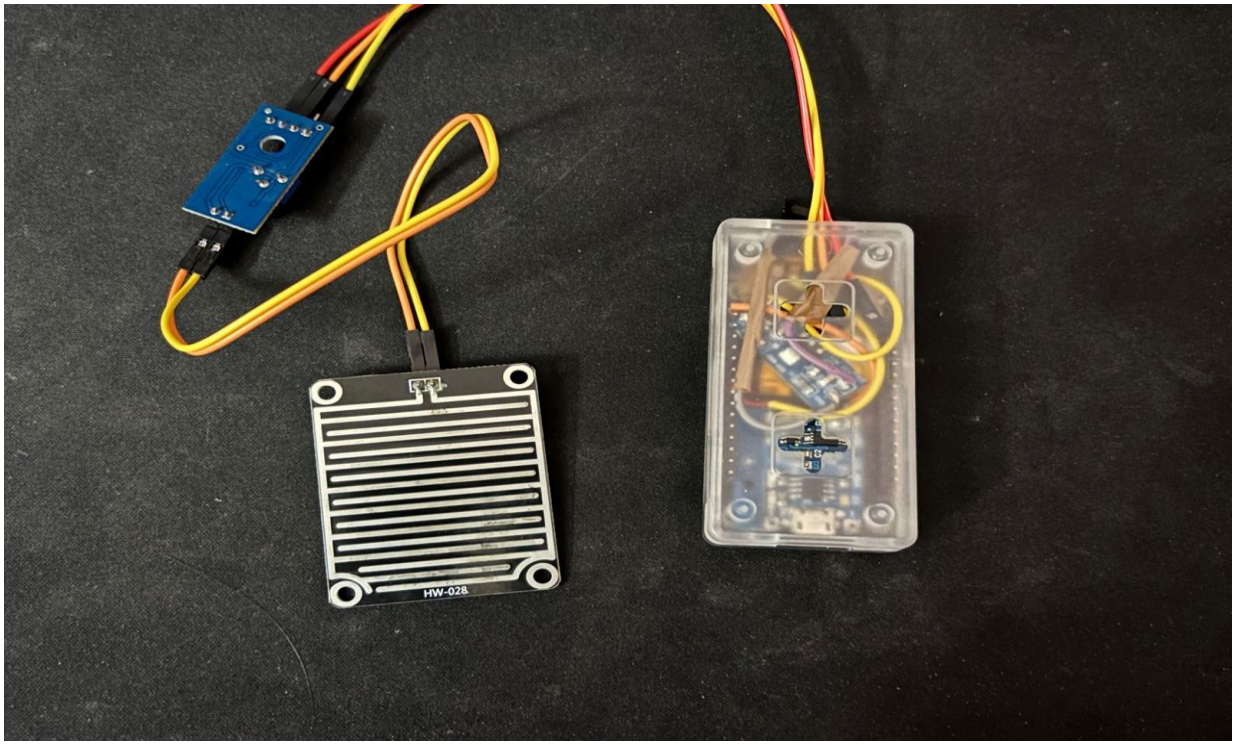
| ITEM | QUANTITY | PRICE |
|---|---|---|
| ESP8266 | 1 | 250₹ |
| WET SENSOR | 1 | 150₹ |
| LI-ION BATTERY | 1 | 100₹ |
| TP4056 MODULE | 1 | 20₹ |
| AMS1117 | 1 | 15₹ |
| SLIDE SWITCH | 1 | 5₹ |
| JUMPER WIRES, BREADBOARD, MISC | As required | 500₹ |
| | | TOTAL: 1040₹ |

**TABLE 2:** The cost of production of the Sensus cruor

The cost of production of the Sensus cruor is 1040₹

# Chapter 7
# FINAL PRODUCT

The result of the project is the product ‘Sensus Cruor: Smart Menstruation Detection System.’ This system aids differently-abled individuals, particularly those with autism spectrum disorder (ASD), in managing menstruation independently. By integrating a wet sensor, esp8266, and a Telegram bot, the system detects menstrual blood and instantly notifies caregivers, ensuring timely assistance while maintaining privacy and dignity. The combination of hardware components and software used, as detailed in previous sections, has been successfully tested to deliver an efficient, real-time alert mechanism. The final product provides a reliable, hygienic, and supportive solution for menstrual hygiene management, enhancing caregiver efficiency and reducing dependency.



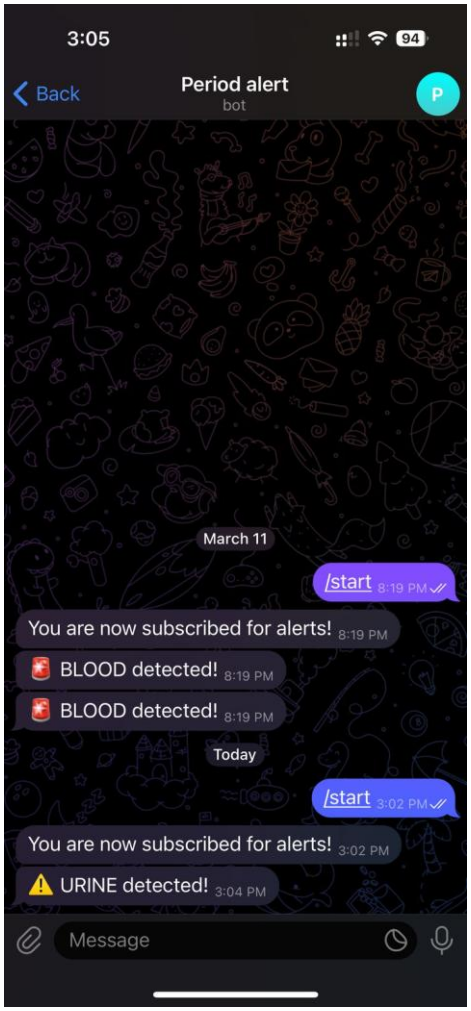

7.1. Sensus cruor

# Chapter 8

# RESULTS &APPLICATIONS

- Support for Differently-Abled Individuals: The *Sensus Cruor* system assists individuals with autism spectrum disorder (ASD) and other disabilities in managing menstruation independently, reducing reliance on caregivers.
- Real-Time Alert Mechanism: The integration of a wet sensor and ESP8266 ensures immediate detection of menstrual blood and urine, triggering real-time notifications to caregivers via Telegram for timely assistance.
- Enhanced Hygiene Management: By detecting both menstrual blood and urine, the system eliminates the need for manual checks, ensuring a more hygienic and comfortable experience for users.
- Privacy and Dignity: Unlike traditional methods, this smart solution maintains the user's privacy while providing necessary support, fostering independence and confidence.
- Healthcare and Caregiving Support: Caregivers, healthcare providers, and special education centers can integrate this system into their support frameworks to improve menstrual and urinary health management for differently-abled individuals.
- IoT-Based Monitoring: The system's IoT integration allows remote monitoring and logging of alerts, making it a scalable solution for care institutions and guardians managing multiple individuals.
- Adaptability for Other Hygiene Monitoring Applications: The core technology can be adapted for other hygiene-related monitoring applications, such as detecting incontinence in elderly care, healthcare environments, and patient management systems.

This was the output of our product, Sensus Cruor, which incorporates the latest technology and software in combination with hardware to create an efficient solution. The product is a part of IoT-based healthcare automation, ensuring real-time monitoring and assistance for differently-abled individuals.




8.1 Alert sent by sensus cruor

# Chapter 9
# CONCLUSION

- Sensus Cruor is an innovative IoT-based solution designed to assist differently-abled individuals, particularly those with ASD, in managing menstruation independently. By integrating sensor technology and real-time alerts, the system ensures timely assistance while maintaining user privacy and dignity.

- Whether for caregivers, healthcare professionals, or individuals requiring support, Sensus Cruor provides an efficient and reliable approach to menstrual hygiene management, reducing dependency and enhancing overall well-being.

- Sensus Cruor incorporates modern technology to automate menstrual detection, offering a practical, hygienic, and supportive solution. This advancement contributes to improved quality of life, ensuring convenience, safety, and dignity for differently-abled individuals.

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